

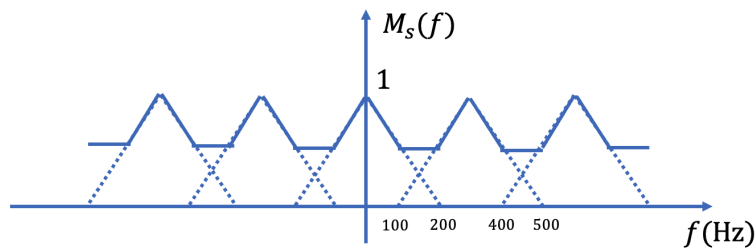
Solutions to Problem Set 2

1. The frequency spectrum of a low-pass signal $m(t)$ is

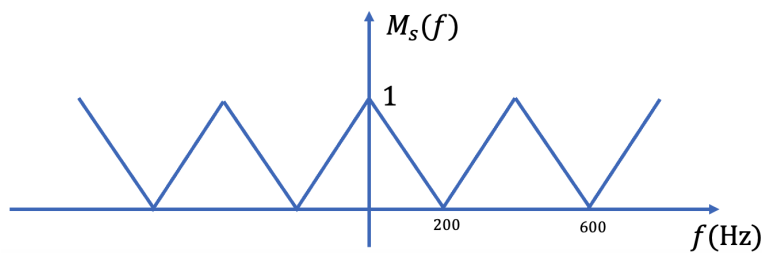
$$M(f) = \begin{cases} 1 - \frac{|f|}{200}, & |f| < 200\text{Hz} \\ 0, & \text{else.} \end{cases}$$

- a) We first sample $m(t)$ with a periodic sequence of unit impulse functions at sampling rate $f_s = 300\text{Hz}$. Draw the frequency spectrum $M_s(f)$ of the sampled signal $m_s(t)$.
- b) If the sampling rate is increased to $f_s = 400\text{Hz}$, please draw $M_s(f)$ again.

Solution. a)



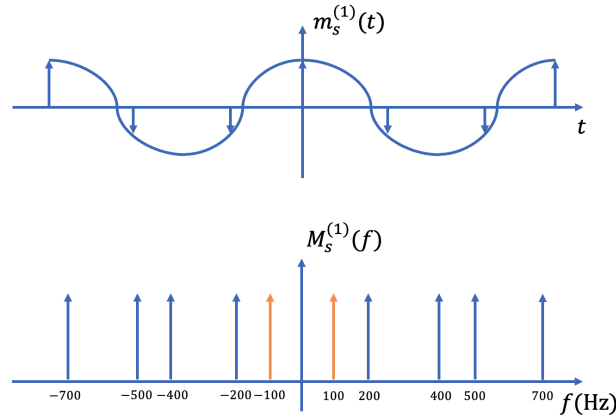
b)



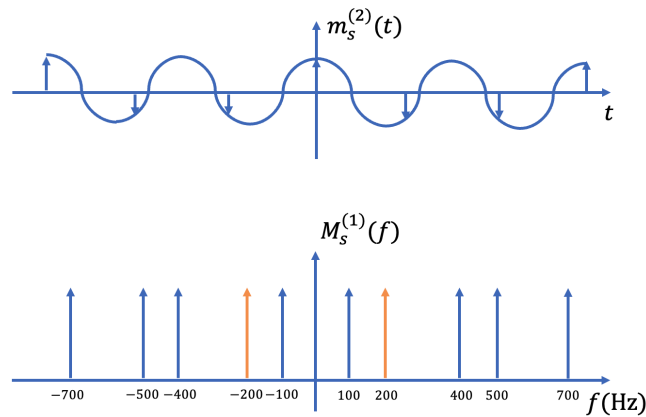
2. In this problem, we examine the aliasing effect in the time domain.

- a) For a sinusoidal signal $m^{(1)}(t) = \cos(200\pi t)$, we conduct ideal sampling at $f_s = 300\text{Hz}$. Draw the sampled signal $m_s^{(1)}(t)$ along with $m^{(1)}(t)$. Also draw the frequency spectrum $M_s^{(1)}(f)$ of the sampled signal.
- b) For another sinusoidal signal $m^{(2)}(t) = \cos(400\pi t)$, the same sampling process is conducted. Draw $m_s^{(2)}(t)$ along with $m^{(2)}(t)$. Also draw the spectrum $M_s^{(2)}(f)$.
- c) Compare $m_s^{(1)}(t)$ and $m_s^{(2)}(t)$ and comment. What signal $m_r(t)$ do we actually reconstruct after applying a low-pass filter at the receiver?

Solution. a)



b)



- c) The two sampled signals $m_s^{(1)}(t)$ and $m_s^{(2)}(t)$ are exactly the same. After applying a low-pass filter at the receiver, we can only reconstruct $m_s^{(1)}(t)$.

3. A sinusoidal signal is uniformly quantized. Suppose the signal fits exactly inside the quantization range of the quantizer, and the dynamic range, i.e. the ratio of the largest value and the smallest value, is 40dB. How many bits are required at least for $\text{SNR}_q \geq 30\text{dB}$ over the dynamic range? (Hint: the normalized effective value D of the sinusoidal wave is $1/\sqrt{2}$)

Solution. Since the dynamic range is 40dB, let the power of the smallest value be $\sigma_{\min}^2$, we have $\sigma_m^2/\sigma_{\min}^2 = 10^4$. It is sufficient for the smallest value to be quantized with $\text{SNR}_{q,\min} \geq 30\text{dB}$, i.e.

$$\begin{aligned} 10\log_{10}\text{SNR}_{q,\min} &= 6.02R + 10\log_{10}\left(\frac{\sigma_{\min}^2}{m_{\max}^2}\right) + 4.77 \\ &= 6.02R + 10\log_{10}\left(\frac{\sigma_m^2}{m_{\max}^2}\right) - 40 + 4.77 \end{aligned}$$

Substituting $D = \sigma_m/m_{\max} = 1/\sqrt{2}$, we have

$$6.02R - 3.01 - 40 + 4.77 \geq 30$$

Therefore, $R \geq 11.33$, and at least 12 bits are required.

4. We are going to conduct the optimal quantization on a Gaussian random variable $X \sim \mathcal{N}(\eta, \sigma^2)$ with only one bit.

a) Find the optimal decision level and representation points.

b) Calculate the expected quantization noise, measured by mean squared error(MSE).

Solution. a) By symmetry of the Gaussian random variable, the optimal decision levels are $\{-\infty, \eta, \infty\}$. For the interval $\{-\infty, \eta\}$, the representation point is

$$\begin{aligned} T_0 &= \frac{\int_{-\infty}^{\eta} xp(x)dx}{\int_{-\infty}^{\eta} p(x)dx} \\ &= \frac{\int_{-\infty}^{\eta} x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\eta)^2}{2\sigma^2}} dx}{0.5} \\ &= \eta + \int_{-\infty}^0 \frac{2}{\sqrt{2\pi\sigma^2}} x e^{-\frac{x^2}{2\sigma^2}} dx \\ &= \eta - \sqrt{\frac{2}{\pi}} \sigma \end{aligned}$$

Similarly, $T_1 = \eta + \sqrt{\frac{2}{\pi}} \sigma$.

b) The expected quantization noise is

$$\begin{aligned} N_q &= \int_{-\infty}^{\eta} (x - T_0)^2 p(x) dx + \int_{\eta}^{\infty} (x - T_1)^2 p(x) dx \\ &= \int_{-\infty}^{\eta} \left(x - \left(\eta - \sqrt{\frac{2}{\pi}} \sigma \right) \right)^2 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\eta)^2}{2\sigma^2}} dx \\ &\quad + \int_{\eta}^{\infty} \left(x - \left(\eta + \sqrt{\frac{2}{\pi}} \sigma \right) \right)^2 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\eta)^2}{2\sigma^2}} dx \\ &= 2 \int_0^{\infty} \left(x - \sqrt{\frac{2}{\pi}} \sigma \right)^2 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} dx \\ &= \sigma^2 \left(1 - \frac{2}{\pi} \right) \end{aligned}$$

5. In this problem, we apply PCM (Pulse Code Modulation) in a voice communication system. It is widely known that human voice has a frequency range of approximately 0.3 to 4 kHz. Therefore in voice communication systems we only care about low-pass signals **below 4kHz**.

- What is the minimum sampling rate f_s for perfect reconstruction below 4kHz?
- In real-world voice signal processing, the total frequency band is divided into sub-bands for separate encoding. In specific, we consider 4 sub-bands, i.e. 0-1kHz, 1-2kHz, 2-3kHz, 3-4kHz. Determine the minimum sampling rate for each of these sub-bands. What is the total sampling rate?
- Given the 8-bit coding protocol below, with input level -314 , compute the PCM code, representation point and noise.

Segment	Segment code			Start Point	Sub-segment code				Quantization interval
	M_2	M_3	M_4		M_5	M_6	M_7	M_8	
0	0	0	0	0	16	8	4	2	2
1	0	0	1	32	16	8	4	2	2
2	0	1	0	64	32	16	8	4	4
3	0	1	1	128	64	32	16	8	8
4	1	0	0	256	128	64	32	16	16
5	1	0	1	512	256	128	64	32	32
6	1	1	0	1024	512	256	128	64	64
7	1	1	1	2048	1024	512	256	128	128

- What is the total bit-rate of this voice communication system? Express the answer in kbps (kilo-bits per second).

Solution. a) According to Nyquist sampling theorem, for low-pass signals below 4kHz, the minimum sampling rate $f_s = 2W = 8\text{kHz}$.

b) For sub-band 0-1kHz, by Nyquist sampling theorem for low-pass signals, $f_{s1} = 2\text{kHz}$. For the other sub-bands, by Nyquist sampling theorem for band-pass signals, $k = 2, 3, 4$, and the minimum sampling rates are 2kHz, 2kHz, 2kHz respectively. The total sampling rate is $2\text{kHz} \times 4 = 8\text{kHz}$. Thus, sub-band division does not introduce extra sampling cost.

c) Because $-314 < 0$, the first bit is '0'. Since 314 is in the segment $[256, 512)$, the segment code is '100'. Subtracting 256 from 314, 58 is left. Note that the sub-segments have same quantization interval 16, and the reconstruction points are 8, 24, 40, 56, 72, 90, ... The closest reconstruction point is 56, corresponding to the sub-segment code '0011', with noise 2. To conclude, the PCM code is '01000011' with noise 2.

d) The total bit-rate is $8\text{kHz} \times 8\text{bits} = 64\text{kbps}$.